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Technical paper

An aircraft engine remaining useful life prediction method based on predictive vector angle minimization and feature fusion gate improved transformer model

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ABSTRACT

Remaining useful life (RUL) prediction is crucial for achieving intelligent and predictive maintenance of aircraft engines. In practical applications, advance prediction values smaller than the true values can prevent serious deferred maintenance accidents. Using asymmetric loss functions directly leads to noticeable accuracy degradation, and existing methods often fail to satisfy accuracy and advance prediction requirements. To address this problem, this paper proposes a novel RUL prediction method based on the Prediction Vector Angle (PVA) minimization and Feature Fusion Gate (FFG) improved Transformer network. Specifically, the FFG is proposed to enhance Transformer prediction accuracy by dynamically fusing global and local features. The concept of PVA is first introduced based on the tilting properties of the RUL descent process. The target of the prediction model is cleverly transformed from error minimization to PVA minimization through the cosine similarity loss function. Various experiments on the CMAPSS dataset demonstrate the effectiveness of the proposed method in achieving high accuracy and advanced prediction. Compared to the state-of-the-art method, RMSE is reduced by at least 2.94 % and Score by 7.00 %. Finally, the PVA minimization mechanism significantly improves long short-term memory and convolutional neural network performance. The proposed method is noteworthy for its superiority and applicability.

1. Introduction

With the continuous advancement of manufacturing technology and the aviation transportation industry, aircraft engine operation and control technology are becoming increasingly complex [1,2]. Reasonably scheduling maintenance plans for essential equipment enhance operational safety and cost-effectiveness [3,4]. Traditional corrective and preventive maintenance methods have become inadequate to meet these demands. As an emerging approach to system reliability assurance, Predictive Health Management (PHM) provides vital benchmarks for intelligent maintenance decisions by predicting the Remaining Useful Life (RUL) [5]. Accurate and reasonable RUL predictions are essential for advancing industrial equipment development in the aerospace, automotive, and battery sectors [6,7]. Therefore, promoting RUL prediction technology is imperative for industrial intelligence transformation [8].

The research on RUL prediction can be generally divided into physics-based and data-driven methods [9,10]. Physics-based methods are difficult to apply due to their complex modeling requirements [11,12]. With the development of artificial intelligence, data-driven methods, especially deep learning methods, are becoming increasingly popular [13,14]. Convolutional Neural Network (CNN) and its variants have unique advantages in dealing with complex signals and disturbances, which enables accurate RUL prediction [15]. However, CNN-based methods are limited by the convolutional kernel sensing field and are not good at capturing contextual long-time features. Recurrent Neural Network (RNN) relies on its internal recurrent structure to process historical information, which confers advantages in RUL prediction [16]. Long Short-term Memory (LSTM) network solves the problem of vanishing gradients in ordinary RNN, but some relevant and important information will still be forgotten [17]. Meanwhile, such methods only support serial computation, which increases the computational burden [18]. As the length of the monitored sequence increases, the drawbacks of traditional CNN, RNN, and their variants become more apparent.

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Nomenclature

Abbreviations

RUL	remaining useful life
PVA	prediction vector angle
PHM	predictive health management
CNN	convolutional neural network
RNN	recurrent neural network
LSTM	long short term memory
MSE	mean squared error
MSLE	mean squared log error
GRU	gated recurrent unit
PVA	prediction vector angle
FFG	feature fusion gating
CMAPSS	commercial modular aero-propulsion system simulation
QTSE	quadratic type squared error
RMSE	root mean square error
Conv1D	one-dimensional convolution
ReLU	rectified linear unit
ACP	self-attention and CNN parallel
ACC	self-attention and CNN contact
FLOPs	floating point operations

Subscripts/superscript

m	multi-head attention
s	self-attention
H	number of attention heads
f	feed-forward layer
k	key matrices
w	window size
t	time
u	utilization weights
sm	smoothing data
j	self-attention head
q	query matrices
v	value matrices
n	columns of sensors signals
T	total length of time
r	regression
e	encoder part
a	advanced
l	lagging

Symbols

Q	query matrices
K	key matrices
X	input matrix
b	bias matrices
$Norm$	normalization
$conv$	convolution operation
h	local features
e	encoder output
y	actual value
$\widehat{y}$	prediction result
x	input sensor data
V	value matrices
W	weight matrices
d	matrix dimension
$ReLU$	rectified linear unit function
O	output matrix
E	error
min	minimum value
max	maximum value
g	global features
u	utilization weights
$\widetilde{u}$	normalized result of u
exp	exponential function
log	logarithmic calculation
Y	target output sample
p	dropout probability
z	output of feature fusion gate
sin	sine function
cos	cosine function
N	batch size
PV	prediction vector
TV	true vector

Greek symbols

γ	smoothing adjustment factor
σ	sigmoid function
λ	adjustment factor
θ	prediction vector angle value

Transformer is a sequence-to-sequence deep learning model that is superior in time series prediction [19]. Compared to CNN, the Transformer can consider the order of the time series through position encoding and obtain better prediction results [20]. Its internal multi-attention mechanism can support parallel computation and has higher computational efficiency than RNN [21]. Transformer has achieved excellent results in large language modeling [22], computer visualization [23], machine translation, etc. However, the ordinary Transformer often ignores local features of the signal, resulting in an unsatisfactory RUL prediction accuracy. Improving the feature extraction ability of the Transformer through the attention mechanism or convolutional layer is the main research direction [21]. Zhu et al. [24] proposed a two-stream convolutional augmented Transformer for predicting the RUL. Zhang et al. [25] designed a dual-aspect self-attention based on the Transformer RUL prediction method. Liu et al. [26] designed a double-attention Transformer, which uses CNN with channel and temporal attention mechanisms for feature learning.

Despite these notable contributions from the above studies, how to improve the accuracy of RUL prediction still needs to be further explored. Consequently, this study presents an analysis of feature usage integrity, which is summarized in Table 1. After analysis, it is evident that neither attention mechanisms nor convolutional layers alone can achieve comprehensive feature extraction. Notably, the gating mechanism effectively enhances the Transformer's feature extraction capability. Xia et al. [27] fused the attention mechanism and the original signal using the gating mechanism, but the extraction of local features is still insufficient. Mo et al. [28] fused the convolutional result and the original signal. Geng et al. [29] fused the convolutional signals at different scales. However, these methods are complex and easy to ignore global features. Proposing a novel mechanism that can comprehensively extract global and local features is the key to improving the RUL prediction accuracy, which is also the primary problem to be addressed in this paper.

On the other hand, most traditional and Transformer-based methods are studied to enhance RUL prediction accuracy. A critical limitation in existing research is the need for more consideration of practical applicability. Predicting the RUL for predictive maintenance must avoid overestimating RUL, which can lead to delays in maintenance and potentially severe accidents [34,35]. While early predictions may incur economic losses, they are generally less harmful than delayed predictions [13,32]. Therefore, alongside traditional accuracy requirements, there is a pressing need to emphasize the performance of

Table 1
Reference analysis on feature usage integrity.

Method	Reference	Temporal feature	Global feature	Local feature	Gating mechanism	Drawbacks
CNN or LSTM based	[15]			√		Ignore temporal information
	[30]		√	√		
	[17]			√		
	[16]	√				Insufficient feature extraction, non-parallelizable
	[31]	√	√			
	[32]	√				
	[3]	√	√			Non-parallelizable
	[14]	√		√		
	[5]	√	√	√		
	[13]	√	√	√		
	[33]	√	√	√		
Transformer based	[21]	√	√			Ignoring local features
	[18]	√	√			
	[19]	√	√			
	[20]	√	√			
	[26]	√	√			
	[24]	√	√	√		Poor feature fusion
	[25]	√	√	√		
	[27]	√	√	√	√	Insufficient local feature
	[28]	√	√	√	√	Complex structure, insufficient global feature
	[29]	√	√	√	√	

early predictions that align with practical operational demands in the RUL prediction process. Obtaining prediction results that are align with practical demands is another critical issue explored in this paper.

To improve the practicality of RUL prediction, customizing the asymmetric loss function is an effective method. The asymmetric loss function improves the probability of early prediction by increasing the loss of larger RUL predictions, forcing the model output to be smaller than the actual value [36]. Divish et al.[37] conducted engine RUL prediction experiments on bidirectional LSTM, deep-learning neural network, and CNN using different loss functions. The experimental results found that the Mean Squared Error (MSE) loss function outperforms most asymmetric loss functions in accuracy and advancement metrics. Therefore, using only an asymmetric loss function without changing the network structure or training method can easily lead to a prediction that is too small to meet the accuracy requirement [38]. Early prediction must be predicated on sufficient accuracy, so the problem cannot be solved using only a simple asymmetric loss function [39]. A few researchers have tentatively solved the problem using combinatorial loss functions. Zhu et al. [24] introduced regularization into MSE to construct an asymmetric loss function and used a parameter to adjust the degree of asymmetry. Liu et al. [40] also applied this loss function. Zhou et al. [39] proposed a loss function consisting of MSE, asymmetric function, and regularization for gas turbine degradation trend prediction. Those methods need to adjust the weights of each component continuously during the training process, which greatly increases model complexity [24,39,40]. The above loss functions are goal-oriented rather than designed based on the specificity of the actual problem, resulting in poor interpretability and practicality. Furthermore, current loss functions are considered too simplistic that early prediction should not commence in all processes. At the initial stages of either non-degradation or early degradation, minimal predictive values can lead to decision-making errors. Constructing a more appropriate loss function based on RUL descent characteristics and mechanisms is another key target of this paper.

In summary, the existing Transformer models fail to integrate global and local features effectively, resulting in unsatisfactory accuracy. The loss function used to achieve prediction advancement lacks interpretability and is poorly considered. To address these problems, this paper innovatively proposes an improved Transformer based on Prediction Vector Angle (PVA) minimization and Feature Fusion Gating (FFG). Various experiments are conducted on the public CMAPSS dataset, and the results confirmed the superiority of the proposed method. The main contributions of this paper are shown below:

(1) **A feature fusion gating mechanism is proposed to enhance the feature extraction capability of the Transformer.** Global features and local features are extracted using a self-attention mechanism and convolutional layer, and then different dimensional features are adaptively blended.

(2) **The concept of prediction vector angle is proposed for the first time in the field of RUL prediction.** Based on the inherent tilting tendency of the degradation process, the training objective is adjusted to the prediction vector angle minimization, which is larger during lagged prediction compared to advanced prediction.

(3) **Training loss calculation is realized using cosine similarity.** The altered loss function has no any adjustable parameter and achieves both prediction accuracy and advancement. The proposed loss function has an improved effect on both CNN and LSTM models and is highly adaptable.

(4) **The proposed model outperforms most existing models, achieving a superior balance between precision and complexity.** The average root mean square error and Score are 7.00 % and 24.08 % lower than the lowest CNN-based or RNN-based methods, 2.94 % and 9.11 % lower than the lowest Transformer-based methods.

The main structure is as follows. Section 2 introduces the principles of the methods used. Section 3 performs RUL prediction and parameter tuning experiments. In Section 4, various comparative experiments are carried out to demonstrate the superiority of the proposed methods. Finally, Section 5 presents conclusions and prospects.

2. Proposed method

In this section, the proposed method based on Prediction Vector Angle (PVA) minimization and Feature Fusion Gate (FFG) is presented in detail for a better understanding. Under this scheme, the Transformer model, feature fusion gate, prediction vector angle, and the framework used are shown below.

2.1. Transformer model

The Transformer is a sequence-to-sequence prediction model achieved through an encoder and decoder. Its hallmark is the utilization of multi-head attention mechanisms to capture long-range dependencies across different time positions. Additionally, it employs residual connections to prevent gradient vanishing or explosion. The Transformer

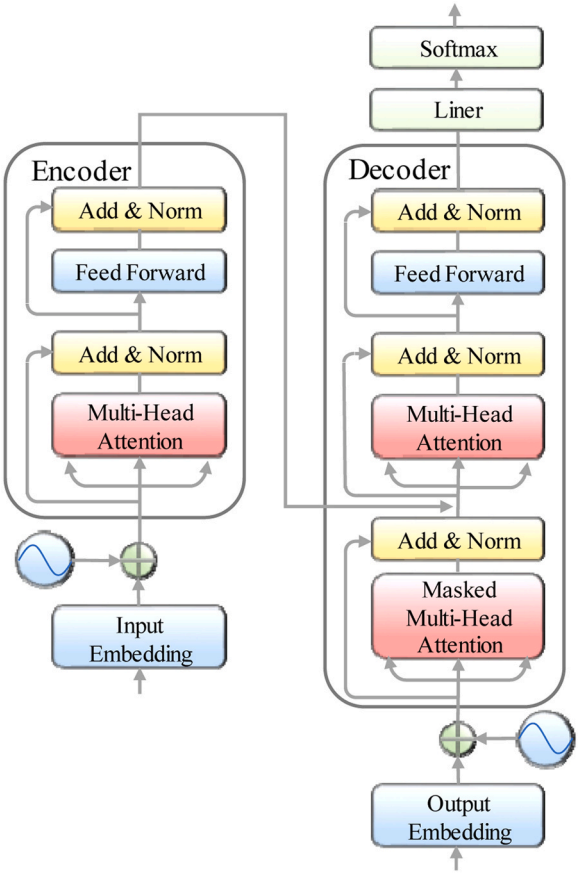


Fig. 1. Structure of Transformer model.

boasts efficient parallel computing capabilities, making it more advantageous than traditional CNN and RNN models in RUL prediction tasks. The structure of the Transformer is illustrated in Fig. 1.

The principles of the encoder part will be described in detail. Firstly, multi-head attention, which consists of multiple heads self-attention functions, is used to extract global features. The output of an attention function is computed as a weighted sum of the values, where the weight assigned to each value is computed by a compatibility function of the query with the corresponding key. The output of the j-th self- attention head is defined as Eq.(1).

$$Attention(Q,K,V)_j = \boldsymbol{Softmax}\left(\frac{Q_j K_j^T}{\sqrt{d^k}}\right)V_j \tag{1}$$

where $Q = X_s W_j^q$, $K = X_s W_j^k$ and $V = X_s W_j^v$ are query, key and value matrices, respectively. W_j^q, W_j^k and W_j^v are weight matrices of self-attention head. X_s is the input feature matrix, and d^k is the dimension of key matrix. Combining multiple self-attention outputs to focus on information from different performance subspaces at different locations. The output is defined as Eq.(2).

$$MultiHead(Q,K,V) = Concat\left(\left\{Attention(Q,K,V)_j\right\}_{j=1}^{H}\right)W_m \tag{2}$$

where W_m is multi-head attention weight matrix, and H is the total number of attention heads.

Subsequently, each multi-head attention module is linked by residuals to prevent model degradation and gradient anomaly, and normalization is used to speed up convergence. The final output X_m of the multi-head attention is defined as Eq.(3).

$$X_m = Norm(X_s + MultiHead(Q,K,V)) \tag{3}$$

where *Norm* stands for normalization.

The Feed-Forward layer performs non-linear computations on the outputs of the multi-head attention. It consists of two linear transformations with a Rectified Linear Unit (ReLU) activation in between, which are applied independently at each time step. The residual calculation is also used to get the final output O_e of the encoder, and the process can be defined as Eq.(4) and Eq.(5).

$$FeedForward(X_m) = W_{f,2}\cdot ReLU\left(W_{f,1}X_m + b_{f,1}\right) + b_{f,2} \tag{4}$$

$$O_e = Norm(X_m + FeedForward(X_m)) \tag{5}$$

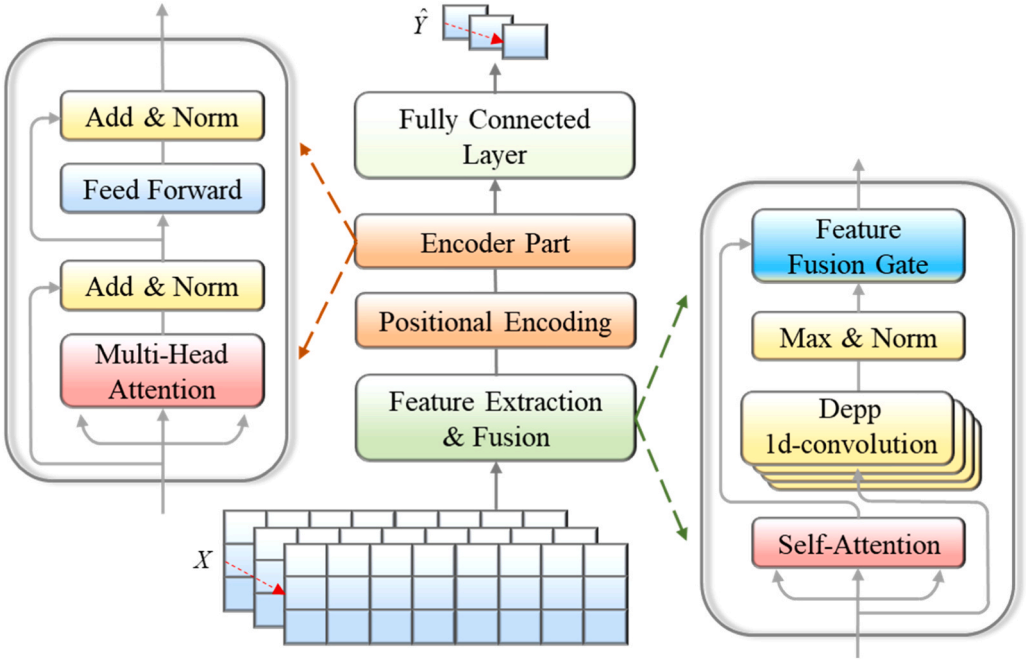


Fig. 2. Structure of FFG-Transformer model.

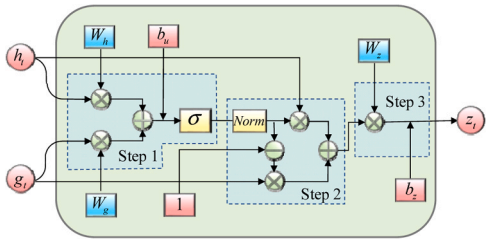


Fig. 3. Structure of FFG.

where $W_{f,1}$, $W_{f,2}$, $b_{f,1}$, and $b_{f,2}$ are weight and bias matrices, respectively.

The encoder part of the Transformer is actually formed by stacking multiple identical modules as described above. The decoder has an additional masked multi-head self-attention mechanism, as well as a Softmax layer at the end to output the probability of the next output result. In this paper, only the encoder part is used in the prediction model, so the decoder is not described in detail.

2.2. FFG-Transformer model

To improve the RUL prediction accuracy of the Transformer encoder part, an improved self-attention and convolutional neural network parallel module based on gating mechanism is designed. The principle of the proposed FFG-Transformer is shown in Fig. 2.

Self-attention mechanism and convolutional neural network features are used to extract global and local features, respectively. The FFG is designed for balancing the global and local features. The principles of the computational flow are described next.

(1) Self-attention and convolution operation

Given an input sequence $X = [X_1 \quad \cdots \quad X_t \quad \cdots \quad X_T]$. The $X_t \in R^{w\times n}$ is the data within n columns of sensors signals at t moment, and w represents the time span. Global features g_t of the input vectors are extracted using the self-attention mechanism, and local features h_t are extracted using a multilayer one-dimensional convolutional (Conv1D) neural network. The process is defined as Eq.(6) and Eq.(7).

$$g_t = Attention(Q, K, V) \tag{6}$$

$$h_t = conv(X_t) \tag{7}$$

where *conv* represents Conv1D operation. The combination of the two can achieve a more comprehensive and accurate feature representation, effectively enhancing the learning ability of the model.

(2) Feature fusion gate

The simple addition of global and local features does not effectively improve the prediction accuracy of the Transformer. Referring to the gating mechanism in LSTM and GRU, a novel mechanism is proposed to adaptively perform feature fusion for features of different dimensions. The difference is that the proposed gating mechanism only involves the input features at the current moment. The realization process can be divided into three steps, as shown in Fig. 3.

Step 1. Utilization weights calculation: Similar to the forgetting and memorizing factor of LSTM training process, the utilization weights u_t of global and local features are controlled by the sigmoid function. The process is shown in Eq.(8).

$$u_t = \sigma\left(W_h h_t + W_g g_t + b_u\right) \tag{8}$$

where W_h, W_g and b_u are trainable parameters.

Step 2. Feature fusion adaptive: The $u_t \in (0,1)$, thus the global and local features are converted into composite features using this utilization weight coefficients. This process can be defined as Eq.(9).

$$z'_t = h_t \otimes u_t + g_i \otimes (1 - u_t) \tag{9}$$

Since u_t is automatically updated based on the loss during the training process, the fusion ratio of the two features can also be seen as adaptive optimal.

Step 3. Linear mapping: Finally, a linear mapping to adapt the dimension of the features to that of the encoder layer, as shown in Eq. (10).

$$z_i = W_z z'_i + b_z \tag{10}$$

where W_z and b_z are trainable parameters.

(3) Position encoding

The position encoding part gives a unique representation of the position by adding a random vector p to the fused feature z, which allows the model to learn temporal information by relative position easily. Normally, the sine and cosine functions with different frequencies are used for position encoding, as shown in Eq.(11).

$$\begin{aligned} p_t^{2s} &= \sin\left(t/10000^{2s/d}\right) \\ p_t^{2s+1} &= \cos\left(t/10000^{2s/d}\right) \end{aligned} \tag{11}$$

where d denotes the matrix length, and $s = 0, 1, \ldots, d-1$ is used to distinguish between odd and even positions of p_t. The relatively large number of 10000 is chosen because trigonometric functions will essentially not have the same wavelength in this range, so the performance of the model does not depend on the choice of this value [24].

The final output of this part is shown in Eq.(12). Temporal order features are learned through Eq.(12) and p_t is fed into the encoding part for further feature extraction.

$$e_t = z_t + p_t \tag{12}$$

(4) Regression layer

Finally, the fully connected regression layer is utilized to map the features of the encoder output e_t into the target to obtain the prediction results $\hat{y}$ of the model.

$$\hat{y}_t = \sigma(W_r e_t) + b_r \tag{13}$$

where W_r and b_r are trainable parameters of the regression layer. Finally, the loss is calculated, and all the parameters are updated according to backpropagation.

2.3. Prediction vector similarity loss function

Advance prediction refers to predicting values $\hat{y}$ smaller than the actual values y, effectively preventing delayed maintenance dates compared to lagging predictions. The mean squared log error function is a traditional asymmetric loss functions, while it is susceptible to outliers and cannot be computed at certain values [39]. In contrast, reference [41] proposes a more robust Quadratic Type Squared Error (QTSE) loss function that utilizes the asymmetry of the exponential function to regulate the prediction results. The QTSE is defined as Eq.(14).

$$QTSE = \frac{1}{T}\sum_{t=1}\left((y_t - \hat{y}_t)^2 \exp(\lambda(y_t - \hat{y}_t))\right) \tag{14}$$

where λ is the adjustment factor, and exp is the exponential function.

The symmetric QTSE is simply constructed by taking the error's absolute value as an input to Eq.(14), which penalizes predictions greater or less than the true value equally. The two functions at $\lambda = 0.3$

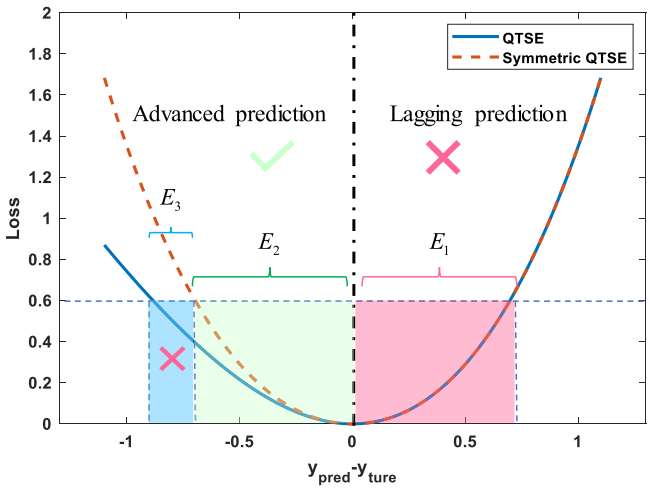


Fig. 4. Characteristics of asymmetric loss function.

are shown in Fig. 4. It is assumed that the loss of the model using the symmetric loss function is maintained at 0.6, and that the prediction errors are distributed on average in the range E_1 and E_2. When an asymmetric loss function is used, the predicted values will tend to be smaller than the true values. However, since the model performance has not been improved enough, the loss will not decrease too much. There will be some prediction errors in the E_3 range, which is not desired.

Therefore, using only the asymmetric loss function would decrease in the prediction accuracy, which is undesirable. In this paper, a concept of minimization of the prediction vector angle is proposed based on the mechanism of the RUL prediction process. The principle of the PVA minimization is shown in Fig. 5(a).

Firstly, assuming that the predicted sequence is $[\hat{y}_1 \quad \hat{y}_2 \quad \cdots \quad \hat{y}_t]$ and the true sequence is $[y_1 \quad y_2 \quad \cdots \quad y_t]$. In the Flight-Cycles-RUL 2-D Cartesian coordinate system, the prediction and the true vector can be defined as Eq.(15) and Eq.(16).

$$PV = (t-(t-1), \hat{y}_t - y_{t-1}) = (1, \hat{y}_t - y_{t-1}) \tag{15}$$

$$TV = (t-(t-1), y_t - y_{t-1}) = (1, y_t - y_{t-1}) \tag{16}$$

Where the t is the moment corresponding to y and $\hat{y}$. The PV and TV are the prediction and true vector, respectively. The PVA is the absolutely value of the angle between PV and TV.

From Fig. 5(b) and (d), the first-order derivative of PVA has poor properties and is not suitable to be used directly as a loss function. This paper introduces the cosine similarity as a loss function. This process is defined as Eq.(17) and Eq.(18).

$$\cos(\theta_i) = \frac{PV \cdot TV}{|PV||TV|} = \frac{1 + \left(\hat{y}_{i,t} - y_{i,t-1}\right)\left(y_{i,t} - y_{i,t-1}\right)}{\sqrt{1 + \left(\hat{y}_{i,t} - y_{i,t-1}\right)^2}\sqrt{1 + \left(y_{i,t} - y_{i,t-1}\right)^2}} \tag{17}$$

$$Loss = \frac{1}{N}\left(\sum_{i=1}^{N}(1 - \cos(\theta_i))\right) \tag{18}$$

where N is the batch size.

When the degradation is not apparent, sufficient prediction accuracy should be ensured to prevent wrong decisions. As shown in Fig. 5(b) and (c), the PVA is symmetric at this point. Lagged and early forecasts do not affect the PVA, so its minimization only means the error is minimized. When degradation occurs, it is necessary to focus on the advance of the forecast. From Fig. 5(d) and (e), it can be seen that the PVA of early prediction at this time will be significantly smaller than the one of the lagged prediction. Therefore, by minimizing the value of PVA, the prediction accuracy and advance can be guaranteed simultaneously. It is worth highlighting that the proposed loss function has no adjustable parameters, which significantly reduces the complexity of the traditional combinatorial loss function.

2.4. Framework

To improve the accuracy and usability of the remaining useful life prediction, this paper innovatively proposes an improved Transformer based on prediction vector angle minimization and feature fusion gating. The main framework of the proposed method is shown in Fig. 6.

The main steps of the proposed methodology are as follows:

(1) Data preprocessing. This includes data collection, normalization, noise reduction, and sample construction.

(2) Model training. Different dimensional features are extracted using a self-attention and convolutional module and then blended using a fusion gate. Subsequently, the Transformer model is trained using the PVA loss function.

(3) Model evaluation. This mainly includes validating the effectiveness of the proposed PVA loss function and feature fusion gates and comparing it with the past state-of-the-art techniques.

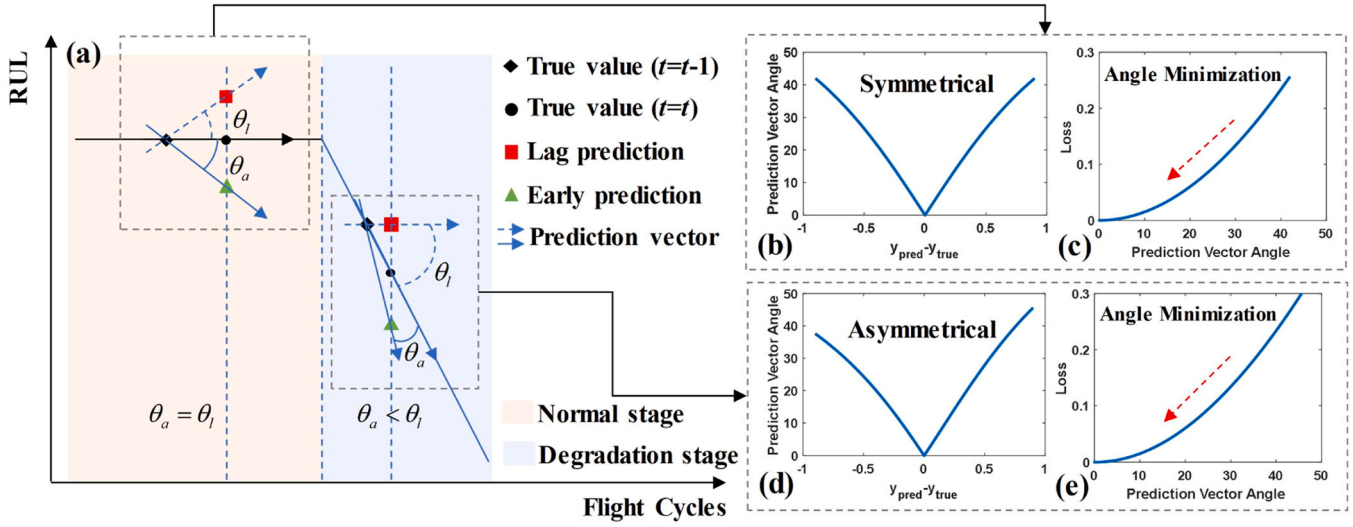


Fig. 5. Principle of PVA minimization: (a) Definition process of PVA; (b), (c) PVA without degradation; (d), (e) PVA in degradation.

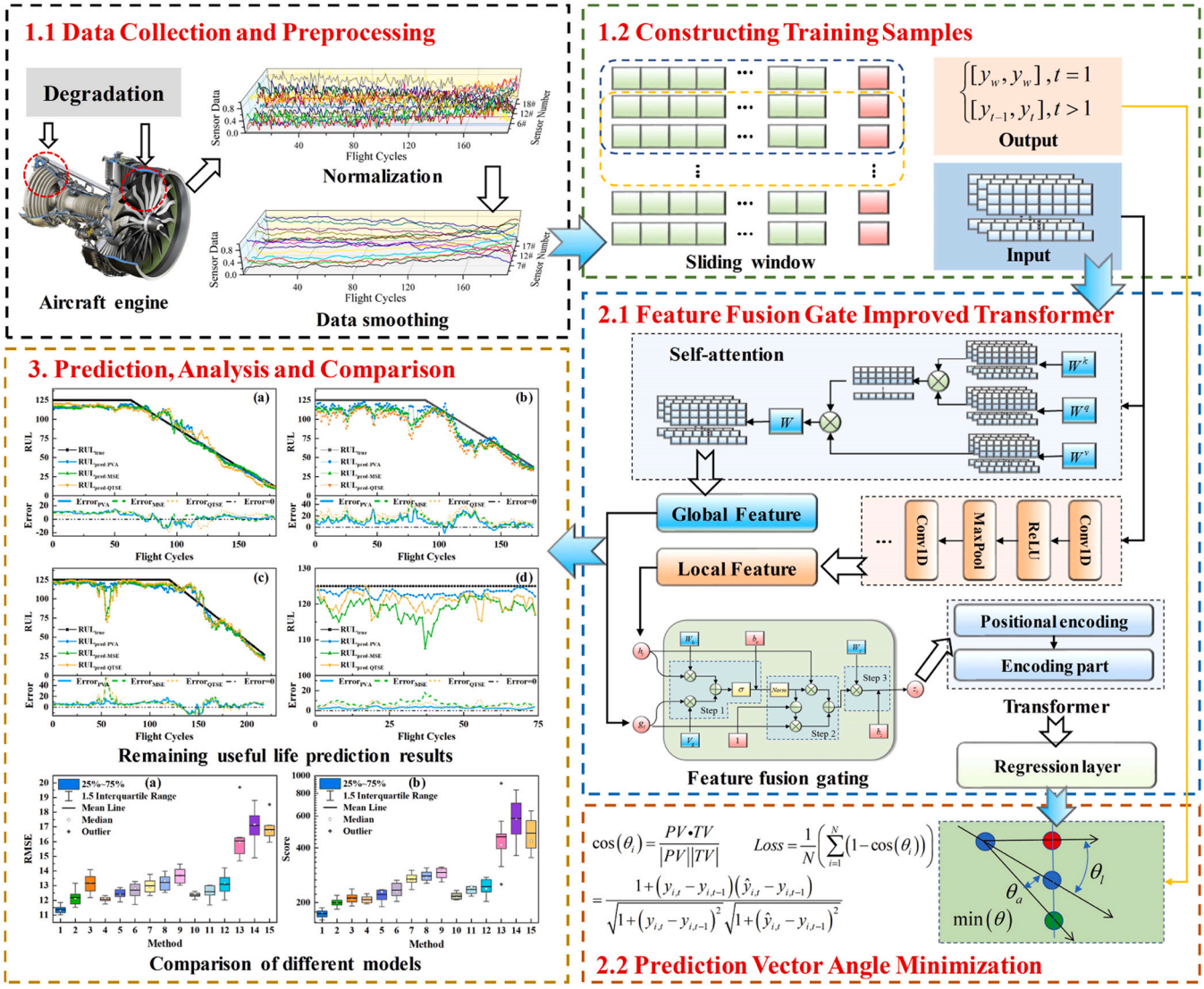


Fig. 6. Framework of the proposed methodology.

Table 2
Structure of the CMAPSS dataset.

	FD001	FD002	FD003	FD004
Train	100	260	100	249
Test	100	259	100	248
Operating conditions	1	6	1	6
Fault model	1	1	2	2

3. RUL prediction experiments

Since actual aircraft engine degradation data are challenging to obtain, RUL prediction experiments will be conducted using public datasets in this section. Specifically, it includes data preprocessing, model parameter tuning, RUL prediction results presentation, and sensitivity analysis.

3.1. Dataset description and processing

3.1.1. Dataset description

The Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) dataset is widely used in RUL predictive model validation, with more than 28 % of studies using it [42]. It consists of four sub-datasets

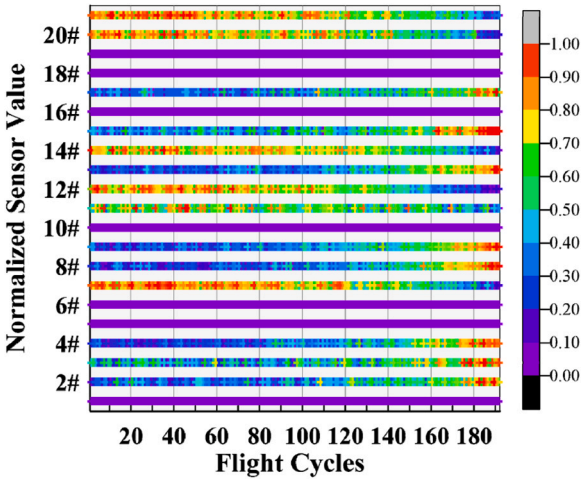


Fig. 7. Trends in normalized data.

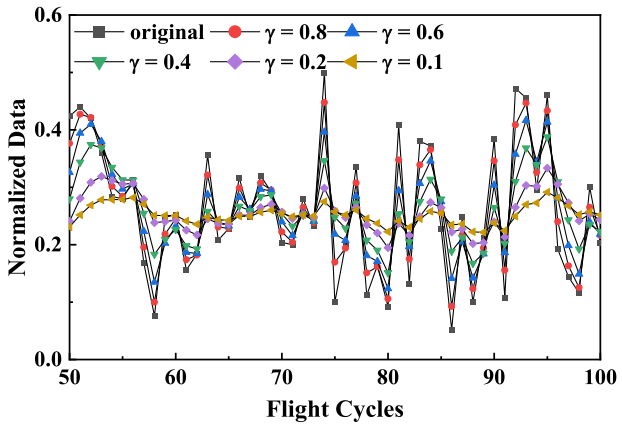


Fig. 8. Results with different smoothing adjustment factor.

(represented by FD001, FD002, FD003, and FD004) with different aircraft engine failure modes and operating conditions, as shown in Table 2. Therefore, CMAPSS is selected to carry out RUL prediction experiments and verify the effectiveness of the proposed method.

The train and test sets for one engine in each sub-dataset are 25-dimensional data. Column 1 is the engine serial number, column 2 is the number of engine cycles, columns 3–5 are the operational settings, and columns 6–26 are the sensor sequence data [43].

3.1.2. Data pre-processing

Data preprocessing is necessary before model training, including normalization and noise reduction. Normalization can accelerate model convergence, which is defined as shown in Eq.(19).

$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)} \tag{19}$$

where max and min represent the maximum and minimum value functions, respectively.

Taking the first engine of FD001 as an example, the normalized data trends are shown in Fig. 7. It can be seen that sensors 1, 5, 6, 10, 16, 18, and 19 have no relationship with the life decline trend, which should be deleted first. Generally, engine performance does not decline in early operation. In reference to existing studies and data characteristics, the upper RUL limit is set to 125 [32,44]. Therefore, RUL normalization is directly defined as dividing the original value by 125.

Noise reduction processing is generally implemented using exponential smoothing, which can improve the model-fitting effect [31]. The definition of exponential smoothing is shown in Eq.(20).

$$x_t^{sm} = \frac{\sum_{i=1}^{t} x_{t-i}^{norm}(1-\gamma)^{i-1}}{\sum_{i=1}^{t}(1-\gamma)^{i-1}} \tag{20}$$

where $\gamma \in (0,1)$ is used as the smoothing adjustment factor. Taking the 2nd sensor data of the first engine in FD001 as an example, the processing results with different γ are shown in Fig. 8. The curve is smoother as γ decreases, but useful information may also be discarded. Therefore, the choice about γ will be determined in through subsequent experiments.

3.1.3. Sample construction

The model input and output samples are constructed using a sliding window approach. The dataset will be disrupted before the model training. To prevent neighboring outputs losing their true decay relationship and to calculate the PVA loss, the output Y_t must be specially changed. The process can be shown in Fig. 9.

The inputs X_t and outputs Y_t at moment t Obtained from Fig. 9 can be defined as shown in Eq.(21) and Eq.(22).

$$X_t = \begin{bmatrix} x_{t-w+1}^1 & x_{t-w+1}^2 & \cdots & x_{t-w+1}^{n-1} & x_{t-w+1}^n \\ \vdots & \vdots & \cdots & \vdots & \vdots \\ x_t^1 & x_t^2 & \cdots & x_t^{n-1} & x_t^n \end{bmatrix} \tag{21}$$

$$Y_t = \begin{cases} [y_w, y_w] & , t = 1 \\ [y_{t-1}, y_t] & , t > 1 \end{cases} \tag{22}$$

Table 3
Details of parameter tuning experiment.

Parameter	Range
Window size	[10,20,30,40]
Minibatch sizes	[10, 20,3 0, 40]
Smoothing factor	[0.1, 0.2, 0.4, 0.6, No smoothing]
Dropout factor	[0, 0.1, 0.2, 0.3]
Weight decay	[0.000001, 0.000005, 0.00001, 0.0001, 0.0005, 0.001]
Learning rate	[0.1, 0.01, 0.001, 0.0001]
Number of convolutional layer	[1–3]
Number of encoder part	[1–3]

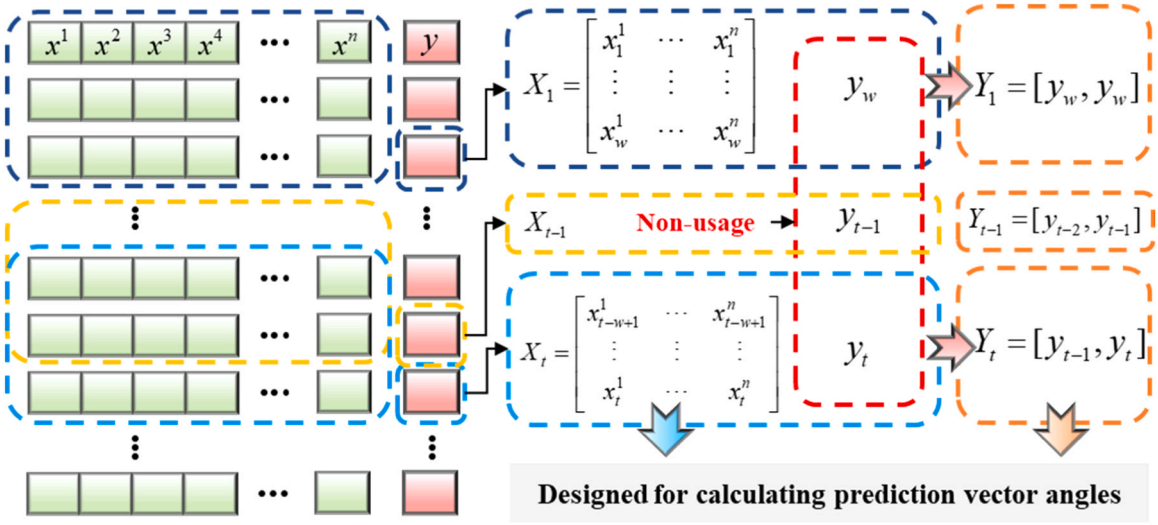


Fig. 9. Sample construction process.

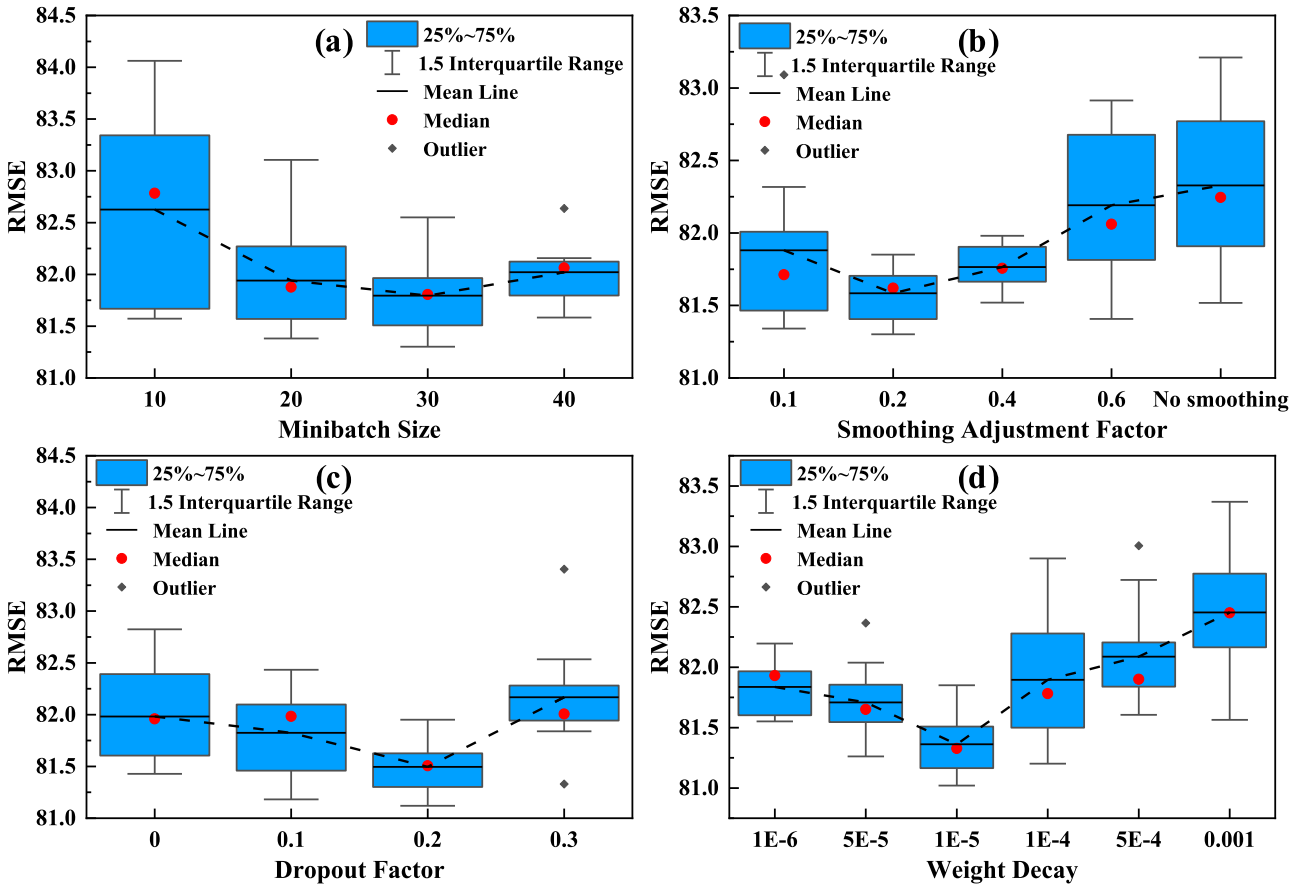


Fig. 10. Results of the tuning experiment: (a) Validation RMSE for different minibatch sizes; (b) Validation RMSE for different smoothing factors; (c) Validation RMSE for different dropout factors; (d) Validation RMSE for different weight decay.

Table 4
Structure of the FFG-Transformer.

Components	Layers	Parameters
Input	Input size	(Total length, 1, 30, 14)
Gating	Self-Attention	heads= 1
	CNN layers	Conv1D(30, 30, kernel size=3, padding=1) ReLU Conv1D(30, 30, kernel size=3, padding=1) ReLU
	Feature Fusion Gate	
RUL Positional	Positional encoding	Dropout, p= 0.2 Normalization layer
Encoder block	Encoder layer	Dropout, p= 0.2 Self-attention, heads= 4 Encoder block, num= 1
Output	Fully connected layer	Output layer, hidden units: 1
Hyperparameters	Learning rate	0.001
	Minibatch size	30
	Weight decay	0.00001
	Max epochs	200
	Smoothing Factor	0.2

Table 5
RMSE and Score of the proposed model.

FD001		FD002		FD003		FD004	
RMSE	Score	RMSE	Score	RMSE	Score	RMSE	Score
11.36	173.89	13.24	714.98	11.80	215.32	15.16	1049.53

where w represents the window size. Unlike conventional methods, the output sample Y has two components. The y_t is the target value corresponding to X_t. The y_{t-1} is introduced to compute the PVA and the model loss.

3.2. Performance evaluation indexes

The Root Mean Square Error (RMSE) and Score function are selected to evaluate the model performance. RMSE is used to evaluate the accuracy of the model prediction. Score penalizes heavily when the predicted value is larger than the actual value and thus is used to evaluate the model's advance prediction ability. The definitions of RMSE and Score are shown in Eq.(23) and Eq.(24), respectively.

$$RMSE = \sqrt{\frac{1}{T}\sum_{t=1}(y_t - \hat{y}_t)^2} \tag{23}$$

$$Score_t = \begin{cases} \exp(-(y_t - \hat{y}_t)/10) - 1, & y_t - \hat{y}_t \le 0 \\ \exp((y_t - \hat{y}_t)/13) - 1, & y_t - \hat{y}_t > 0 \end{cases}$$
$$Score = \sum_{t=1}^{T} Score_t \tag{24}$$

3.3. Model hyperparameter and realization details

The model parameters have a significant effect on the prediction results. To obtain the optimal configuration, crucial hyperparameters are adjusted with a single control variable in this section. The training data is divided into a training set and a validation set according to 8:2,

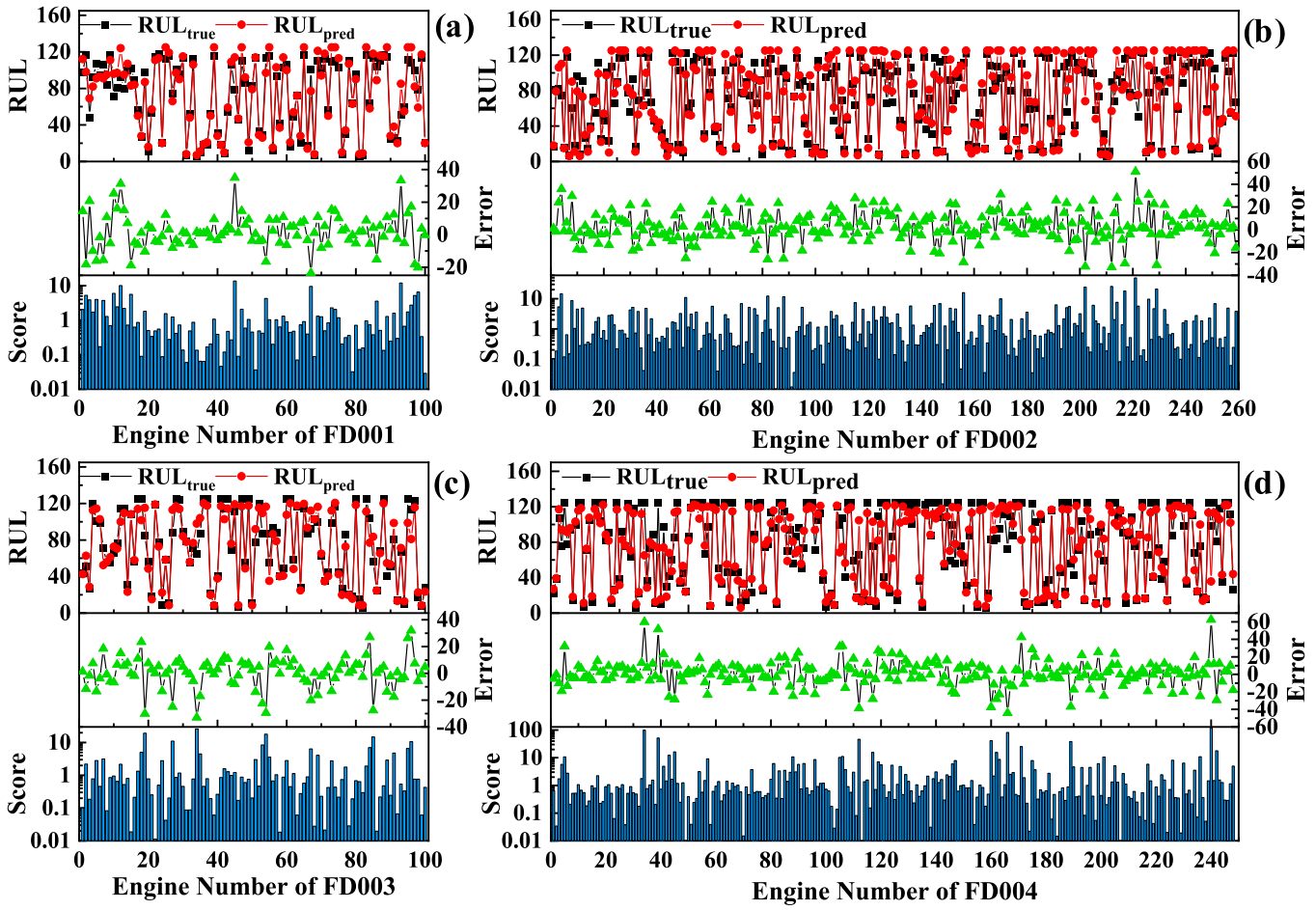


Fig. 11. RUL prediction results of the proposed model: (a) Prediction results of FD001; (b) Prediction results of FD002; (c) Prediction results of FD003; (d) Prediction error and Score of FD004.

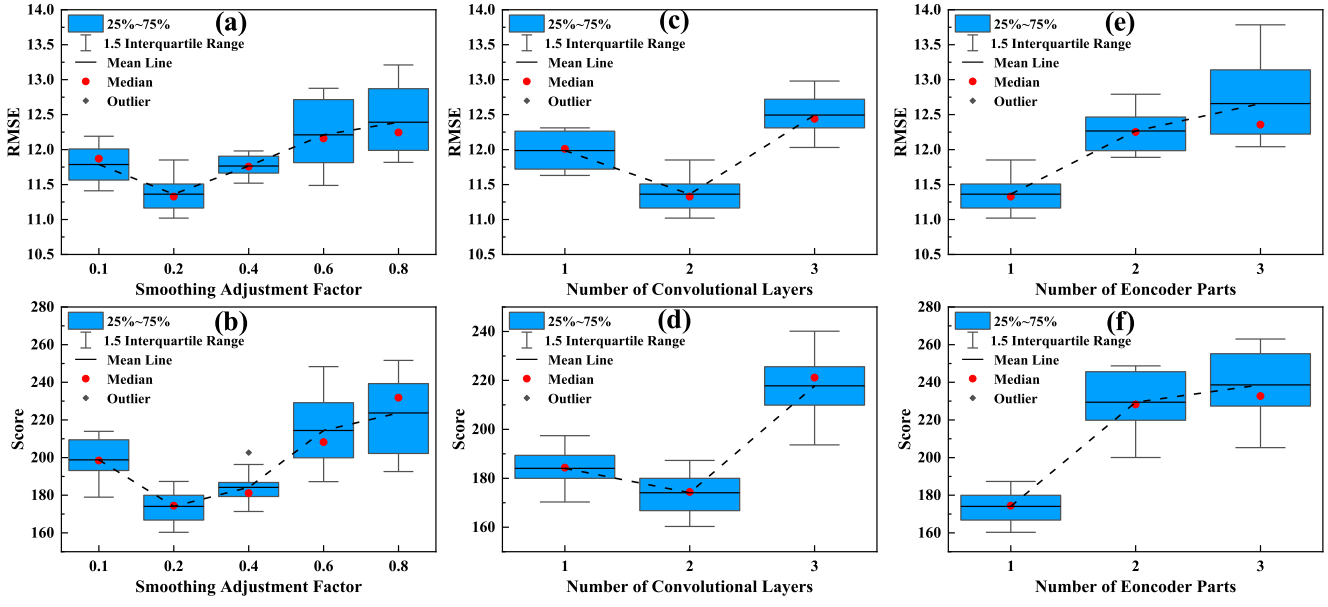


Fig. 12. Parameter sensitivity analysis results: (a), (b) Test RMSE and Score for different smoothing factors; (c), (d) Test RMSE and Score for different number of convolutional layers; (e), (f) Test RMSE and Score for different number of encoder parts.

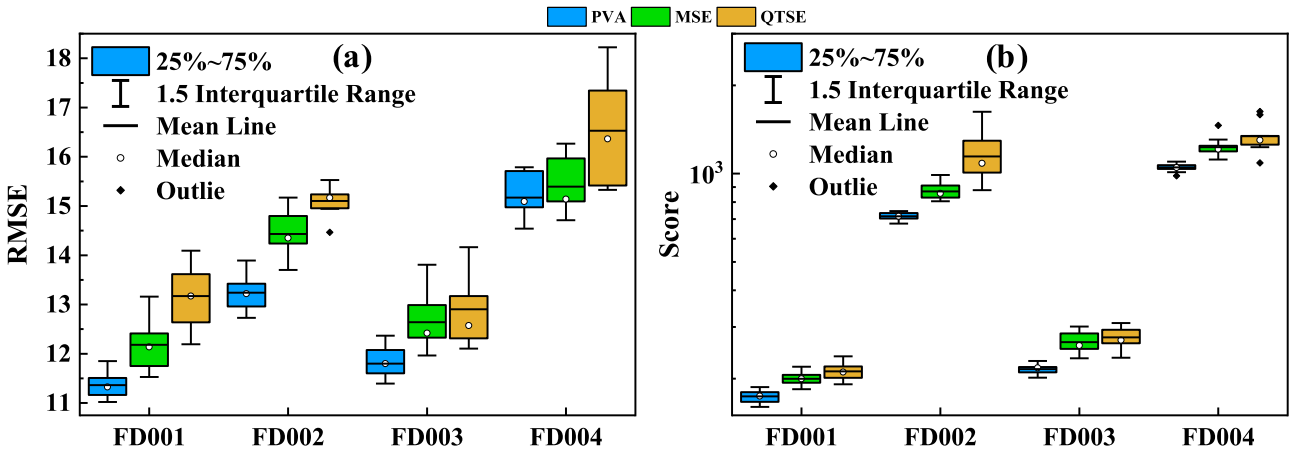


Fig. 13. Results with different loss functions.

Table 6
Comparison of metrics and improvements of different loss functions.

	Metrics	PVA	MSE	Improvement	QTSE	Improvement
FD001	RMSE	**11.36**	12.16	6.58 %	13.07	13.08 %
	Score	**173.89**	199.83	12.98 %	211.92	17.95 %
FD002	RMSE	**13.24**	14.43	8.24 %	15.1	12.34 %
	Score	**714.98**	867.78	17.61 %	1124.68	36.43 %
FD003	RMSE	**11.80**	12.63	6.60 %	12.89	8.46 %
	Score	**215.32**	265.58	18.93 %	275.77	21.92 %
FD004	RMSE	**15.16**	15.39	1.43 %	16.5	8.06 %
	Score	**1049.53**	1227.99	14.53 %	1322.88	21.26 %
Average	RMSE	**12.89**	13.66	5.57 %	14.39	10.42 %
	Score	**538.43**	640.30	15.91 %	733.81	26.63 %

and the validation RMSE is utilized for tuning. The adjustable parameters and ranges are shown in Table 3.

These parameters are adjusted in sequence so that when one parameter is adjusted, the other parameters select the optimal or maximum value. Taking FD001 as an example, the results of the main hyperparameter tuning experiments are shown in Fig. 10. Other parameter adjustment processes are similar and will not be repeated.

Batch size refers to the frequency of model updates, and experiments show that 30 is optimal. When exponential smoothing is not used, the data contains a lot of noise, resulting in insufficient training accuracy. The prediction accuracy improves as the smoothing factor decreases and reaches the optimum at 0.2, while too small leads to loss of information. The neuron dropout and weight decay are regularization operations to prevent overfitting, and the optimal dropout rate and weight decay are 0.2 and 0.00001, respectively. After experiments, the structure of the proposed PVA-FFG-Transformer model is shown in Table 4.

Finally, all experiments are based on PyCharm 2023.1.3, and the program is written using Python 3.10. Neural network models are trained and tested based on torch 2.01. The hardware platform is 13th Inter (R) Core (TM) i5–13400 processor with a basic frequency of 2.50 GHZ and RAM 32 GB.

3.4. RUL prediction results

RUL prediction experiments are launched on each of the four CMAPSS datasets using the model in Table 4 and the PVA loss function. Each dataset is tested ten times and the prediction error for the last cycle of each test engine is recorded. The average of RMSE and Score are used to evaluate the prediction performance and the results are shown in Table 5.

The prediction results of the optimal model for all the test sets are shown in Fig. 10. As shown in Table 5 and Fig. 11, the proposed model fits the RUL prediction curves well on all four datasets and has low RMSE and Score.

3.5. Parameter sensitivity analysis

To validate the effectiveness of the proposed model, the sensitivity of parameters to final testing outcomes is investigated in this section. The adjustable parameters are also shown in Table 3. Model testing is first conducted with smoothing factors, and the corresponding test metrics are shown in Fig. 12 (a) and (b). The $\gamma = 0.2$ yielded the optimal results, consistent with conclusions drawn from the validation set. This section will focus on the effect of the number of convolutional layers and encoder parts, and the other three parameters illustrated in Fig. 10 will not be repeated.

Keeping the above optimal parameters, the number of convolutional layers is selected as 1, 2, and 3, respectively. The test results are shown in Fig. 12 (c) and (d). Convolutional layers are the key structure for extracting local features, and too small a number will lead to insufficient feature extraction, while too large a number will cause overfitting. The number of encoder parts are also selected as 1, 2 and 3, and the experimental results are shown in Fig. 12 (e) and (f). It can be seen that the model performs best when one encoder part is used, and too many encoder parts will cause overfitting and a decrease in test accuracy. The sensitivity experiments show that each parameter in the proposed method is important and relates to the ability to extract degraded features.

4. Comparative experiments

This section conducts various comparison experiments to verify the effectiveness of the proposed PVA minimization loss function and FFG mechanism. Comparisons will also be made with state-of-the-art

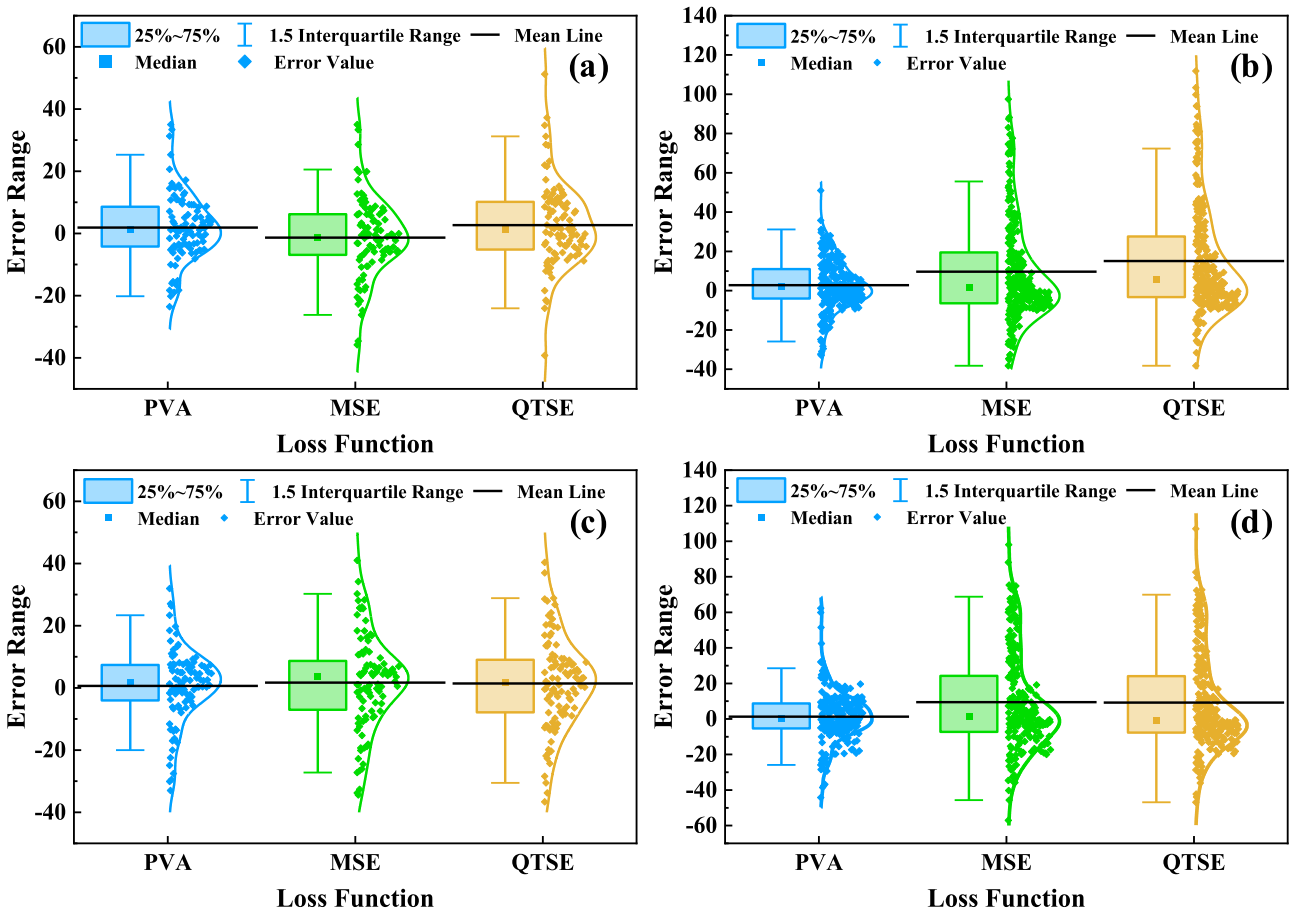


Fig. 14. Error distribution of the proposed model on four datasets: (a) Error distribution on FD001; (b) Error distribution on FD002; (c) Error distribution on FD003; (d) Error distribution on FD004.

methods to demonstrate the superiority of the proposed method.

4.1. Comparative experiments between PVA and traditional loss function

In this section, comparative experiments on different loss functions are carried out to highlight the superiority of the loss function based on the PVA minimization. PVA, MSE, and QTSE are used as loss functions to unfold the prediction under the same network structure, respectively. The value of λ in QTSE function is always set to 1. Each method is run 10 times and the final results are shown in Fig. 13.

The proposed PVA loss function achieves the best prediction results on all four datasets, with the lowest mean and median prediction errors. Meanwhile, the proposed method has the narrowest 1.5-fold quartile, proving it has the highest stability. The average values of RMSE and Score are selected as the comparative indexes shown in Table 6. Meanwhile, Table 6 demonstrates the prediction improvement of the PVA function compared to the traditional loss function.

It is obvious from Table 6 that the proposed PVA loss function is more suitable for the RUL prediction task. The early prediction performance is significantly better than the normal symmetric loss function while obtaining accuracy improvement. It is calculated to reduce 5.57 % and 15.91 % on average in RMSE and Score compared to MSE. Meanwhile, PVA is reduced by 10.42 % and 26.63 % on average on RMSE and Score compared to QTSE. This is because the asymmetry of PVA is extracted from the degradation mechanism without forcing the output to be too small. It can also be seen from the comparison of MSE and QTSE that using only the asymmetric loss function causes a decrease in accuracy, which is in line with the conclusions of the reference [24] and [37]. Next, the effect of the proposed loss function on preventing lagged prediction is discussed in more detail using the FD001 test set. Fig. 14 illustrates the error distribution for the FD001 dataset, where the error represents the actual value minus the predicted value.

It is evident from Fig. 14 that the mean values of the prediction errors of the proposed method are all greater than 0 and are the smallest compared to the other methods. This indicates that the proposed method obtains the highest accuracy of early prediction values. The prediction effect of MSE is in the middle, while the mean error and error distribution are not regular on different datasets. For the asymmetric QTSE, the average error is always greater than zero, but the average error is the largest and the error distribution is the widest. This indicates that the asymmetric loss function leads to unstable prediction results and decreased accuracy. The proposed PVA loss function achieves optimal predictive performance by extracting insights from the essence of the degradation process and benefiting from specific adjustments during training.

Although RMSE and Score only need to be calculated using the prediction error of the last cycle, a comparison of the prediction effects over the entire run can observe whether the prediction methods are effective at any given moment. Fig. 15 shows the prediction process and error for one random engine in the four datasets. The proposed PVA loss function possesses the minimum prediction error and stable early prediction capability in both the healthy stage and the life decline stage of the engine.

4.2. Comparative experiments with different structures

In this section, the impact of different model structures on the RUL prediction results will be discussed. In addition to the three models in

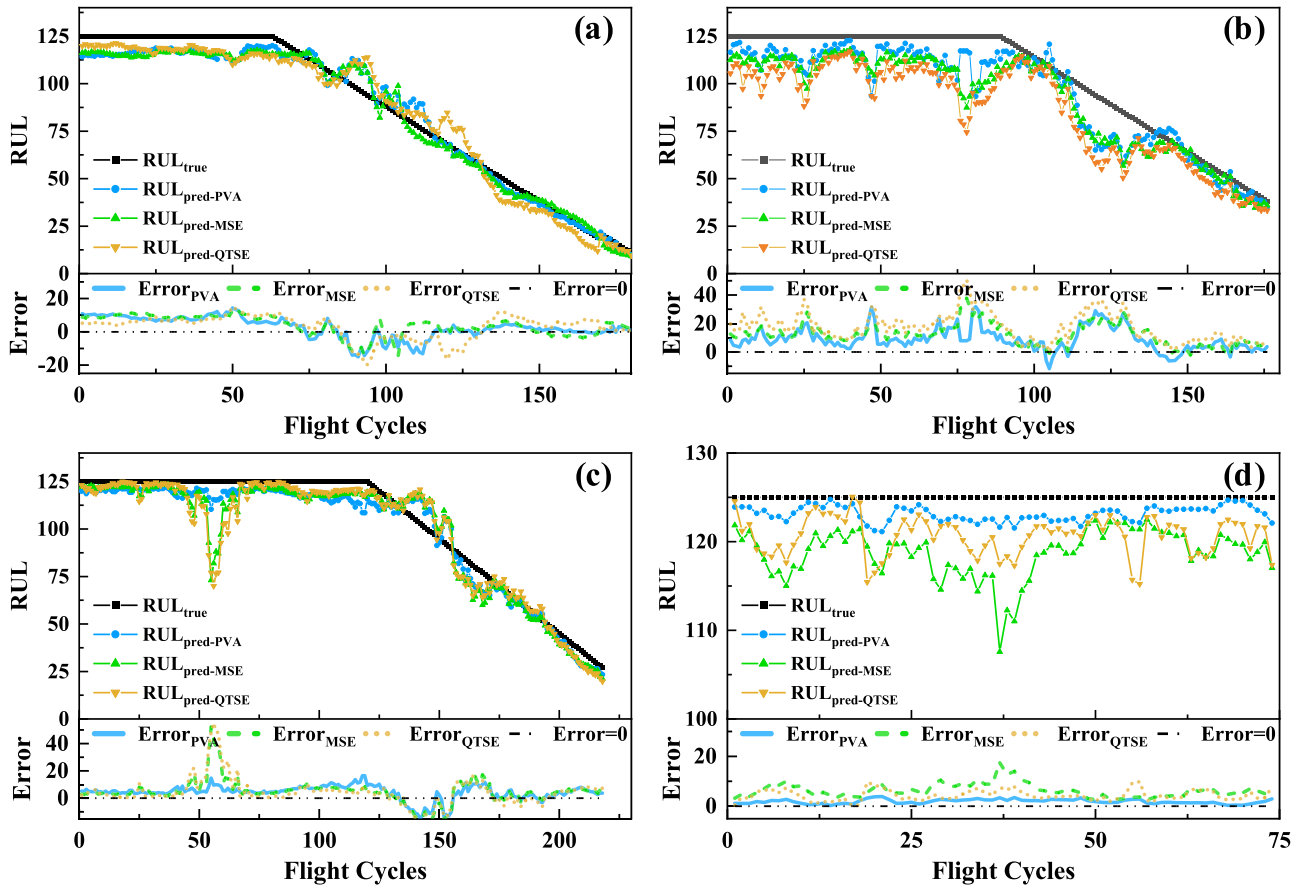


Fig. 15. RUL prediction results for four engines: (a) Prediction results of unit #100 in FD001; (b) Prediction results of unit #62 in FD002; (c) Prediction results of unit #100 in FD003; (d) Prediction results of unit #55 in FD004.

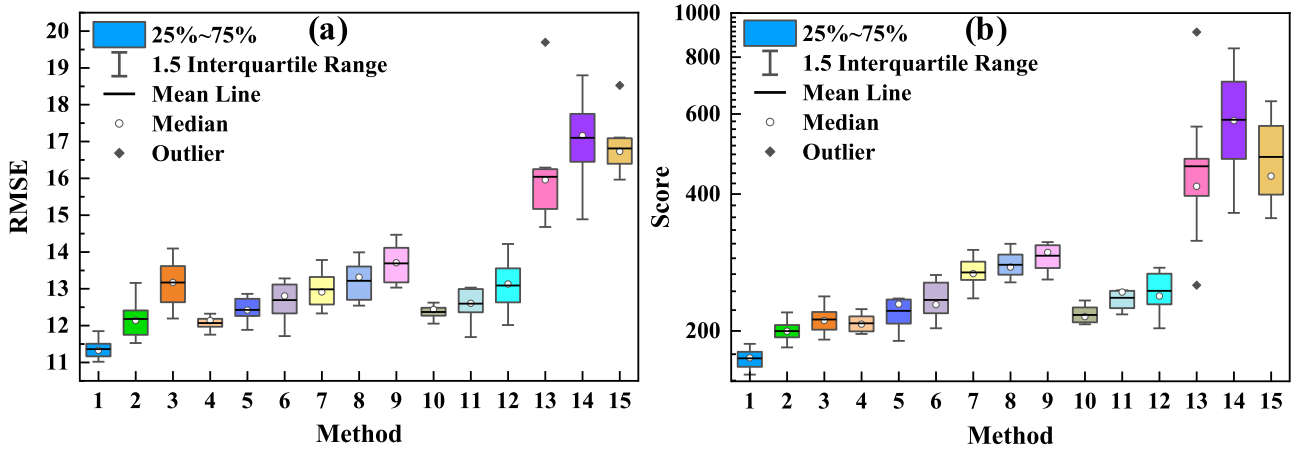


Fig. 16. Prediction results of different Transformer models: (a) RMSE of different Transformer models; (b) Score of different Transformer models.

Section 4.1, 12 different models are built, and experiments are conducted using the FD001 dataset. Among them, the ACP-Transformer refers to the parallel structure of the attention mechanism and CNN, equivalent to the direct addition of the extracted features from both. ACC-Transformer is the tandem structure of the attention mechanism and the CNN, which uses attention to extract the features first and then uses the CNN. The remaining model directly uses either the attention or CNN improved Transformer. Ten experiments were conducted for each model to make the comparison more reasonable, and the metrics of different models are shown in Fig. 16. Methods 1–15 correspond to each of the fifteen models, and the correspondence is shown in Table 7. Meanwhile, Table 7 shows the average RMSE and Score for the different model structures and the percentage improvement of the proposed method compared to other models.

According to Fig. 16 and Table 7, the proposed PVA-FFG-Transformer model performs the best among the 15 comparative experiments. The MSE-FFG-Transformer model outperforms the other Transformer structures using the MSE or QTSE loss function, which indicates the superiority of the proposed FFG structure. The proposed PVA loss function can improve the effect of different Transformer structures, which reflects the practicality of the PVA function.

Table 7
Comparison of metrics and improvements of different structures.

Method		RMSE	Improvement	Score	Improvement
1	PVA-FFG-Transformer	11.36	/	173.89	/
2	MSE-FFG-Transformer	12.16	6.58 %	199.83	12.98 %
3	QTSE-FFG-Transformer	13.07	13.74 %	211.92	17.95 %
4	PVA-ACP-Transformer	12.07	5.88 %	207.93	16.37 %
5	MSE-ACP-Transformer	12.43	8.61 %	221.49	21.49 %
6	QTSE-ACP-Transformer	12.69	10.48 %	233.95	25.67 %
7	PVA-ACC-Transformer	12.99	12.55 %	268.96	35.35 %
8	MSE-ACC-Transformer	13.21	14.00 %	279.63	37.81 %
9	QTSE-ACC-Transformer	13.69	17.02 %	292.80	40.61 %
10	PVA-CNN-Transformer	12.37	8.16 %	216.93	19.84 %
11	MSE-CNN-Transformer	12.60	9.84 %	236.59	26.50 %
12	QTSE-CNN-Transformer	13.09	13.22 %	244.95	29.01 %
13	PVA-Attention-Transformer	16.04	29.18 %	460.33	62.22 %
14	MSE-Attention-Transformer	17.10	33.57 %	582.65	70.16 %
15	QTSE-Attention-Transformer	16.81	32.42 %	482.84	63.99 %

Table 8
Structure of LSTM and CNN.

Deep Learning Architecture	Parameters
LSTM	LSTM, hidden units: 128
	ReLU
	Dropout, p= 0.2
	LSTM, hidden units: 128
	ReLU
	Dropout, p= 0.2
	Linear, Hidden units: 32
	ReLU
	Linear, Hidden units: 1
CNN	Conv1D(14, 32, kernel size=3, padding=1)
	ReLU
	MaxPool(kernel size=2, stride=2)
	Conv1D(32, 64, kernel size=3, padding=1)
	ReLU
	MaxPool(kernel size=2, stride=2)
	Conv1D(64, 128, kernel size=3, padding=1)
	ReLU
	MaxPool(kernel size=2, stride=2)
	Flatten layer()
	Linear, Hidden units: 128
	Linear, Hidden units: 1

4.3. Improvement effect of PVA function on traditional models

To further validate the adaptability of the proposed PVA, prediction experiments should be performed on traditional models. LSTM represents dynamic neural networks, and CNN is a typical model that does not consider time series. These two networks can represent different kinds of existing neural network models. The structure and parameters of the two models are shown in Table 8.

Next, PVA, MSE and QTSE are used as loss functions on LSTM and CNN model. Ten experiments are performed on the four datasets, and the RMSE and Score metrics are shown in Fig. 17.

The following conclusions can be drawn from Fig. 17. First, the proposed PVA loss function has a certain improvement effect on both CNN and LSTM models, reflecting strong adaptability. The prediction accuracy and advance are stable and better than the traditional MSE and QTSE loss functions. In some cases, the asymmetric QTSE loss function outperforms the MSE loss function. However, its performance is unstable, and its effects fluctuate dramatically with the change in the dataset and model. The average value of each experiment is used as the final index, and the results are shown in Table 9 and Table 10.

For the LSTM model, the proposed PVA loss function reduces the RMSE and Score by 2.55 % and 11.75 % on average compared to the MSE loss function. Compared to the QTSE loss function, it reduces the RMSE and Score by 9.21 % and 10.41 % on average, respectively. For the CNN model, the proposed PVA loss function reduces the RMSE and Score by 2.41 % and 13.86 % on average compared to the MSE loss function. Compared to the QTSE loss function, it reduces the RMSE and Score by 3.01 % and 11.95 % on average, respectively. The comparison results demonstrate the significance of performing the loss function construction from the mechanism rather than using the traditional MSE or asymmetric loss function.

4.4. Compare with traditional models

To demonstrate the superiority of the proposed method over the state-of-the-art models, comparisons are conducted across the last four years of RUL prediction algorithms. The comparison models include the traditional models and the improved Transformer-based models. First, the comparison of the RMSE and Score for the general models is shown in Table 11.

Compared to advanced algorithms proposed in the last four years, the proposed method has the lowest average RMSE and Score on all four datasets, which are 7.00 % and 24.08 % lower than the past minimum, respectively. The RMSE on FD001 deteriorated by 2.9 % than the reference [17], but the Score is significantly reduced by 13.9 %. The RMSE on FD003 is 18.11 % worse compared to the reference [33], and the Score is improved by 7.98 %. Meanwhile, the proposed method has the lowest RMSE and Score in FD002 and FD004, which have more complex operating conditions. Subsequently, the comparison of the proposed model with the existing improved Transformer models is shown in Table 12.

The proposed PVA-FFG-Transformer model still achieves the best average metrics on the four datasets. The average RMSE and Score values are 2.94 % and 9.11 % lower than the past minimum. Notably, the proposed method performs best on the FD002 and FD004 datasets, which reflects that the proposed model has a more powerful feature extraction capability. For the FD001 dataset, the proposed model has the smallest Score metric, which indicates that the model has superior advance prediction capability. From Table 11 and Table 12 show that the Transformer-based methods are generally better than the ordinary methods, which also shows the necessity of improving the Transformer in this paper.

4.5. Comparison of model complexity

Model parameters are related to training speed and application cost, which is also one of the metrics to focus on. Two indicators are often applied to evaluate the overall computation quantity: the number of parameters and the number of floating point operations (FLOPs). However, only some references have provided these two indicators, and only four advanced models are compared in this section. The comparison results are shown in Table 13.

Table 13 shows that the proposed method better balances the prediction effect and model complexity. Compared with BiGRU-TSA, PAOL-Transformer, and GCU-Transformer, the proposed model has better prediction accuracy and the smallest number of model parameters. The FLOPs are much smaller than the four comparison models. Although the proposed method increases the number of parameters by 11.21 % compared to DA-Transformer, the FLOPs are reduced by 85.35 %.

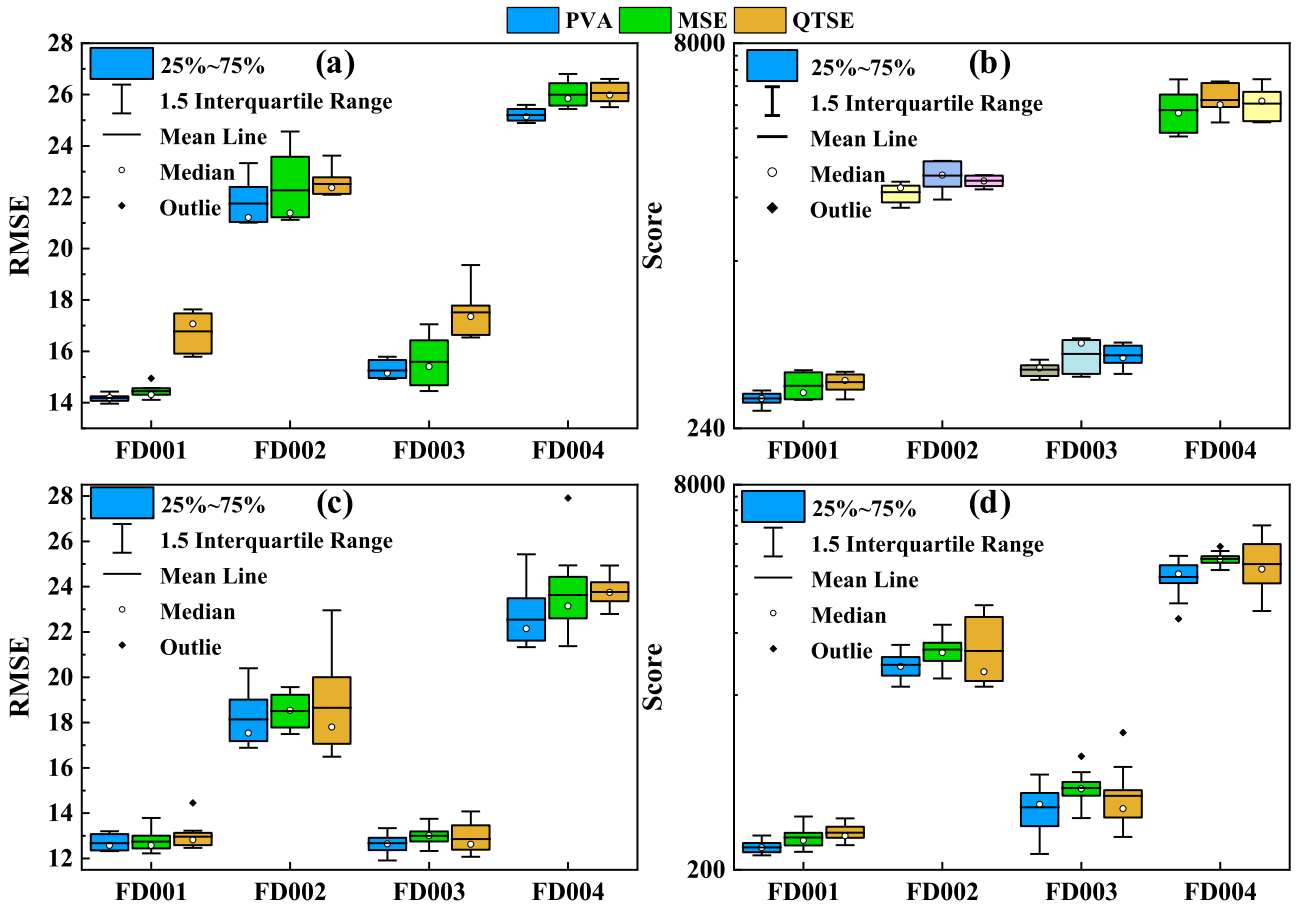


Fig. 17. Prediction results of LSTM or CNN models: (a) RMSE of different LSTM models; (b) Score of different LSTM models; (c) RMSE of different CNN models; (d) Score of different CNN models.

Table 9
Result of different LSTM models.

		PVA-LSTM	MSE-LSTM	Improvement	QTSE-LSTM	Improvement
FD001	RMSE	**14.18**	14.45	1.85 %	16.78	15.47 %
	Score	**314.96**	353.35	10.86 %	365.67	13.87 %
FD002	RMSE	**21.75**	22.26	2.29 %	22.52	3.42 %
	Score	**2061.23**	2398.06	14.05 %	2287.53	9.89 %
FD003	RMSE	**15.25**	15.59	2.16 %	17.51	12.91 %
	Score	**409.7**	472.69	13.33 %	466.28	12.13 %
FD004	RMSE	**25.2**	25.99	3.04 %	26.06	3.29 %
	Score	**4349.21**	4766.80	8.76 %	4614.39	5.75 %
Average	RMSE	**19.78**	20.325	2.55 %	21.54	9.21 %
	Score	**1783.78**	1997.72	11.75 %	1933.47	10.41 %

Table 10
Result of different CNN models.

		PVA-CNN	MSE-CNN	Improvement	QTSE-CNN	Improvement
FD001	RMSE	**12.68**	12.74	0.51 %	12.97	2.23 %
	Score	**247.93**	272.73	9.09 %	285.41	13.13 %
FD002	RMSE	**18.14**	18.51	1.99 %	18.75	3.27 %
	Score	**1423.65**	1648.94	13.66 %	1628.11	12.56 %
FD003	RMSE	**12.68**	13	2.52 %	12.86	1.41 %
	Score	**363.87**	437.61	16.85 %	406.36	10.46 %
FD004	RMSE	**22.55**	23.63	4.61 %	23.76	5.11 %
	Score	**3305.54**	3926.66	15.82 %	3740.77	11.63 %
Average	RMSE	**16.51**	16.97	2.41 %	17.09	3.01 %
	Score	**1335.29**	1571.49	13.86 %	1515.16	11.95 %

Table 11
Comparison with traditional models.

Method	Year	Index	FD001	FD002	FD003	FD004	Average
Proposed Method	2024	RMSE	11.36	**13.24**	11.80	**15.16**	**12.89**
		Score	**173.89**	**714.98**	**215.32**	**1049.53**	**538.43**
DBA-DCRN[17]	2024	RMSE	**11.04**	15.31	11.81	18.96	14.28
		Score	201.94	1084.07	238.05	2209.38	933.36
CATA-TCN[5]	2024	RMSE	12.80	17.61	13.16	21.04	16.15
		Score	234.31	1361.23	290.63	2303.42	1047.40
DA-LSTM[31]	2024	RMSE	12.62	13.22	13.34	16.25	13.86
		Score	263	842	360	1372	709.25
DTW-GPR[1]	2024	RMSE	12.02	17.38	12.70	17.56	14.92
		Score	212.59	1714.30	394.25	2509.30	1207.61
MCDAN[3]	2024	RMSE	/	13.11	/	14.60	13.86
		Score	/	769	/	1082	925.50
$C^2D^2M^2$[2]	2023	RMSE	12.32	20.81	15.32	22.43	17.72
		Score	221	3062	369	3118	1692.50
DBRNN-SLF[16]	2023	RMSE	17.97	19.81	19.18	20.4	19.34
		SCORE	458.89	2064.39	658.12	2869.67	1512.77
SARN[45]	2023	RMSE	13.84	18.92	11.68	17.74	15.55
		Score	/	/	/	/	/
AE-MSEN[44]	2022	RMSE	14.57	20.18	15.58	23.72	18.51
		Score	342.38	1781.81	1231.50	3698.01	1763.43
BiGRU-TSA[46]	2022	RMSE	12.56	18.94	12.45	20.47	16.10
		Score	213.35	2264.13	232.86	3610.34	1580.17
BiGRU-AS[43]	2021	RMSE	13.68	20.81	15.53	27.31	24.06
		Score	284	2454	428	4708	3581
FCLCNN-LSTM[33]	2021	RMSE	11.17	/	**9.99**	/	10.58
		Score	204	/	234	/	219
MCLSTM[32]	2021	RMSE	13.71	/	/	23.81	18.76
		Score	315	/	/	4826	2570.50
MS-MFM[47]	2021	RMSE	/	/	15.07	22.22	18.645
		Score	/	/	609.97	2641.05	1625.51

Table 12
Comparison with different Transformer models.

Method	Year	Index	FD001	FD002	FD003	FD004	Average
Proposed Method	2024	RMSE	11.36	**13.24**	11.80	**15.16**	**12.89**
		Score	**173.89**	**714.98**	215.32	**1049.53**	**538.43**
TATFA-Transformer[21]	2024	RMSE	12.21	15.07	11.23	18.81	14.33
		Score	261.50	1359.70	210.21	2506.35	1084.44
MHT[6]	2024	RMSE	11.92	13.7	**10.63**	17.73	13.50
		Score	215.2	746.7	**150.5**	1572	671.1
RC- transformers[24]	2024	RMSE	**11.12**	13.40	11.02	17.56	13.28
		Score	189.01	918.26	152.73	1109.56	592.39
IMDSSN[20]	2023	RMSE	13.32	19.83	13.92	21.88	17.24
		Score	329.00	3394.25	431.34	3566.14	1930.18
PAOL-Transformer[18]	2023	RMSE	12.49	21.63	12.66	23.86	17.66
		Score	257.71	1692.59	274.15	3163.41	1346.97
GA-transformers[19]	2023	RMSE	11.63	15.99	11.35	20.15	14.78
		Score	215	1133	228	2672	1062
DA- transformers[26]	2022	RMSE	12.25	17.08	13.39	19.86	15.65
		Score	198	1575	290	1741	951
DST- transformers[25]	2022	RMSE	11.43	15.25	11.32	18.36	14.09
		Score	203.15	924.56	154.92	1490.72	693.34
GCU-Transformer[28]	2021	RMSE	11.27	22.81	11.42	24.86	17.59
		Score	N/A	N/A	N/A	N/A	N/A
Transformer	N/A	RMSE	12.92	22.64	13.04	24.81	18.35
		Score	276.27	1960.25	284.41	3892.24	1603.29

Table 13
Comparison of different model parametric quantities.

Model Parameter	Proposed method 1.29E5	BIGRU-TSA[46] 2.82E6	PAOL-Transformer[18] 2.60E5	GCU-Transformer[28] 1.78E6	DA- Transformer[26] 1.16E5
Improvement	/	95.42 %	50.38 %	27.52 %	−11.21 %
FLOPs	1.09E6	1.68E8	/	6.32E7	7.44E6
Improvement	/	99.35 %	/	98.27 %	85.35 %

Meanwhile, the RMSE and Score are reduced by 7.27 % and 12.17 %, respectively. This is because the proposed method uses only the most common Transformer model, enhancing the extraction of global and local features through FFG, which also improves by introducing the PVA loss function. Therefore, the proposed method is more advantageous than only improving the model structure.

5. Conclusion

Aircraft engines inevitably continue to degrade due to the harsh operating environment. Predictive health management can effectively reduce unplanned maintenance and downtime, improving the safety and economics of aircraft operations. Remaining useful life prediction is a key technology to achieve this goal, but improving the usability of the prediction results is a long-term challenge. Addressing this problem and practical needs, this paper proposes an improved Transformer based on prediction vector angle (PVA) minimization and feature fusion gating (FFG). The main conclusions of this paper are as follows:

(1) The proposed feature fusion gate can extract engine degradation features effectively. The proposed gating mechanism can obtain higher and more stable prediction accuracy than using the self-attention mechanism and convolutional layer ground in parallel or using only one of the algorithms.
(2) The proposed loss function based on prediction vector angle minimization can significantly improve prediction accuracy and advancement. Compared to MSE and QTSE, the proposed loss function averages a 5.57 % and 15.91 % reduction in RMSE and a 10.42 % and 26.63 % reduction in Score, respectively.
(3) The proposed PVA is highly adaptable and improves traditional CNN and LSTM models. The LSTM models' average RMSE is reduced by at least 2.55 %, and the average Score by at least 10.41 %, and the CNN models' average RMSE is reduced by at least 2.41 % and the average Score by at least 11.95 %.
(4) The proposed model performs best on the CMAPSS dataset. The average RMSE and Score values are 7.00 % and 24.08 % lower than the lowest CNN or RNN-based methods in the past, 2.94 % and 9.11 % lower than the lowest Transformer-based methods. Meanwhile, the proposed method achieves an optimal balance of accuracy and model complexity.

In summary, the proposed method is important for health management and intelligent equipment operation. The method is promising to be applied in life prediction studies of batteries, automobiles, and tools. However, the volatility of the prediction process is relatively high, which makes it difficult to apply the results directly. In the future, we will try to improve the monotonicity of the prediction results from a mechanism perspective. Meanwhile, we will focus on how the methodology can guide the implementation of actual maintenance strategies.

CRediT authorship contribution statement

Zhenhua Long: Validation, Methodology. **Zhihao Zhou:** Writing – original draft, Methodology, Formal analysis, Conceptualization, Writing – review & editing. **Mingling Bai:** Formal analysis, Conceptualization. **Ruidong Wang:** Visualization, Software. **Daren Yu:** Methodology, Funding acquisition. **Jinfu Liu:** Methodology, Investigation, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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